

CRUD GETX DENGAN API

1. Pengenalan API

Mengapa ada API?

Sebelum kita menuju pembahasan lebih lanjut, sebaiknya kita tahu dulu apa alasan dibuat sebuah API.

Semakin banyaknya penggunaan aplikasi yang tersedia pada berbagai platform, membuat tiap — tiap platform memerlukan adanya integrasi data. Dimana nantinya data pada web dengan data pada aplikasi mobile harus sama agar tidak terjadi duplikasi data pada dua platform yang berbeda.

Tujuan Pembuatan API

Berdasarkan alasan diatas, maka dibuatlah sebuah API yang bertujuan untuk mempermudah proses pertukaran data antar aplikasi yang terintegrasi serta untuk mempercepat proses development dengan menyediakan function secara terpisah sehingga developer tidak perlu membuat fitur yang serupa.

Apa itu REST API?

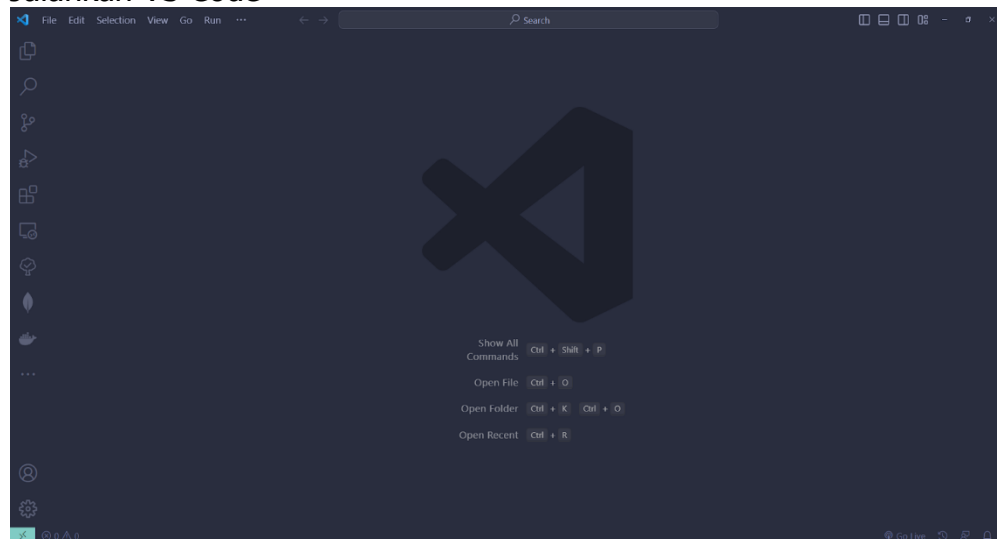
Jika API digunakan untuk proses pertukaran data antar aplikasi yang terintegrasi, maka

REST API adalah salah satu teknologi API yang sering kita gunakan ketika akan mengembangkan sebuah Web APIs.

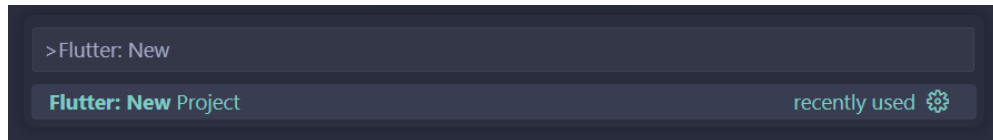
REST API sendiri bersifat stateless. Dimana setiap kali request harus menyertakan semua data dan parameter dengan lengkap ketika mengakses suatu endpoint. Pada arsitektur REST, REST server menyediakan resources (sumber daya/data) dan REST client mengakses dan menampilkan resource tersebut untuk penggunaan selanjutnya. Setiap resource diidentifikasi oleh URIs (Universal Resource Identifiers) atau global ID. Resource tersebut direpresentasikan dalam bentuk format teks, JSON atau XML. Pada umumnya formatnya menggunakan JSON dan XML.

2. Membuat Project Flutter CRUD API

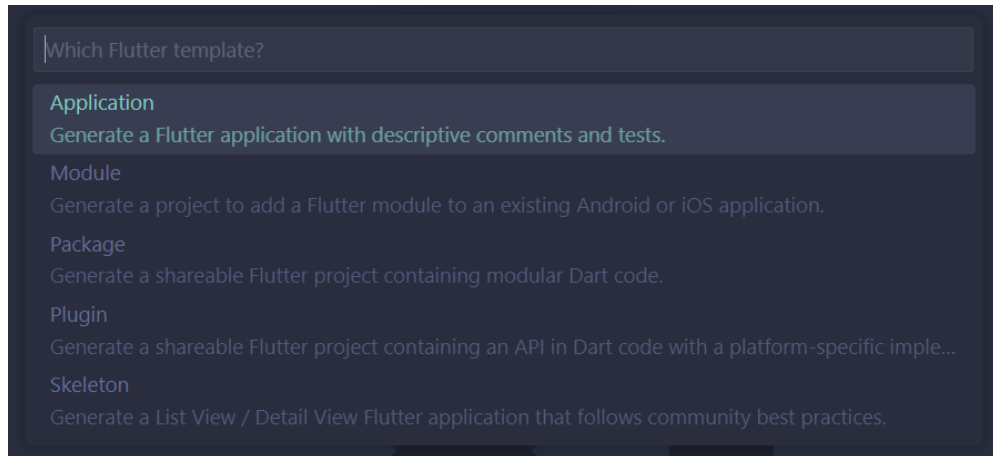
1. Jalankan VS Code



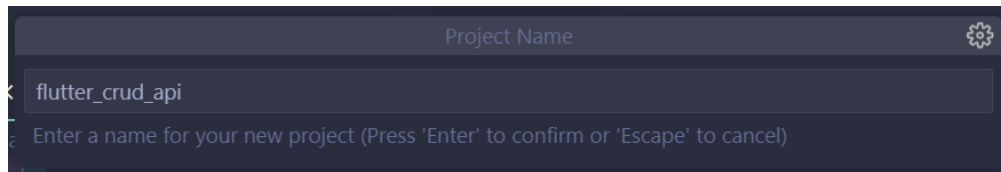
b. Ketik CTRL + SHIFT + P kemudian ketik dan pilih 'Flutter: New Project'.



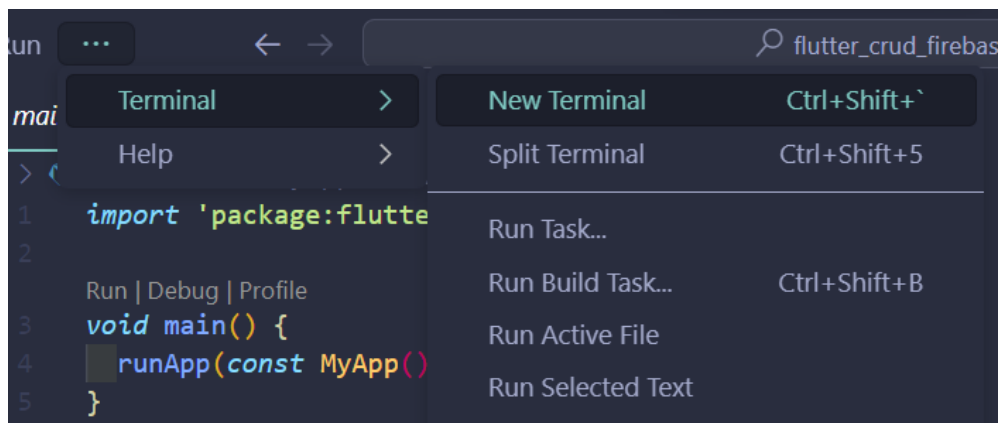
- c. Pilih Application, lalu pilih lokasi penyimpanan project yang akan digunakan.



- d. Berikan nama project, disini penulis memberikan nama 'flutter_crud_api'.



- e. Selanjutnya jalankan perintah 'get init' pada terminal VS Code



- f. Selanjutnya pilih 'GetX Pattern' dan pilih 'Yes', maka akan menjadi seperti berikut ini:

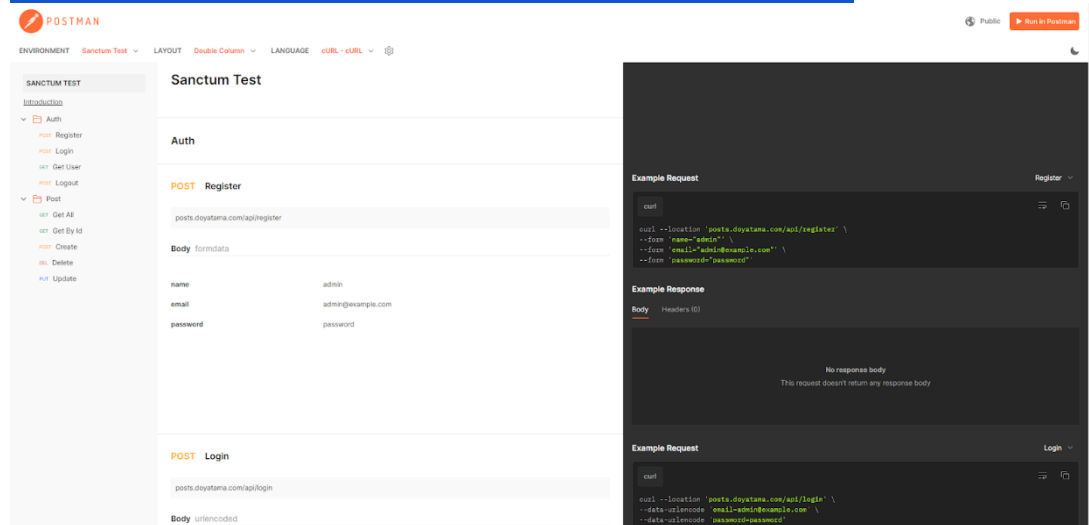
```
PS F:\Project\Palatihan\Flutter\flutter_crud_api> get init
? Which architecture do you want to use? (Use arrow keys)
> GetX Pattern (by Kauê)
  CLEAN (by Arktekko)
? Your lib folder is not empty. Are you sure you want to overwrite your application?
WARNING: This action is irreversible (Use arrow keys)
> Yes!
  No
  ✓ 'Package: get installed!
  ✓ File: main.dart created successfully at path: lib\main.dart
  ✓ File: home_controller.dart created successfully at path: lib\app\modules\home\controllers\home_controller.dart
  ✓ File: home_view.dart created successfully at path: lib\app\modules\home\views\home_view.dart
  ✓ File: home_binding.dart created successfully at path: lib\app\modules\home\bindings\home_binding.dart
  ✓ File: app_routes.dart created successfully at path: lib\app\routes\app_routes.dart
  ✓ File: app_pages.dart created successfully at path: lib\app\routes\app_pages.dart
  ✓ home route created successfully.
  ✓ Home page created successfully.
  ✓ GetX Pattern structure successfully generated.

Running 'flutter pub get' ...
```

- g. Setelah selesai selanjutnya bisa menonaktifkan kode berikut ini pada file 'test/widget_test.dart'

```
testWidgets('Counter increments smoke test', (WidgetTester tester) async {
  // Build our app and trigger a frame.
  // await tester.pumpWidget(const MyApp());
})
```

3. Akses dokumenter API yang sudah disediakan pada halaman berikut ini:
<https://documenter.getpostman.com/view/14200360/2s946k5qdT>



Pada halaman ini berisi banyak API, misalnya digunakan untuk login, register, logout, dan juga untuk CRUD posting.

4. Selanjutnya mengintegrasikan pada project flutter yang sudah dibuat sebelumnya. Langkah awal yang dilakukan adalah melakukan instalasi package yang dibutuhkan dan dilanjutkan membuat beberapa widget yang akan digunakan beberapa kali.

```
flutter pub add flutter_svg intl http bot_toast get_storage google_fonts
```

5. Atur halaman main seperti berikut ini untuk melakukan init GetStorage (media penyimpanan lokal yang disediakan oleh get) serta melakukan konfigurasi utama untuk toast / modal.

```
Future<void> main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  await GetStorage.init();  
  Intl.defaultLocale = 'id_ID';  
  runApp(  
    GetMaterialApp(  
      title: "Application",  
      initialRoute: AppPages.INITIAL,  
      getPages: AppPages.routes,  
      builder: BotToastInit(),  
      navigatorObservers: [BotToastNavigatorObserver()],  
    ),  
  );  
}
```

6. Selanjutnya buat beberapa folder untuk menyimpan services, utils dan widgets pada folder lib/app.
- a. services
Digunakan untuk membuat penerimaan data dari API yang sudah disiapkan, misalnya mengambil data API untuk login, register, menampilkan data post dsb.
 - b. utils
Digunakan untuk menyimpan variabel global seperti warna dan setting rute API.
 - c. widgets
Digunakan untuk menyimpan potongan-potongan widget yang akan digunakan kembali, misalnya untuk toast, alert, dsb.
7. Buat file konfigurasi untuk API pada folder utils dengan nama api.dart

```
class SharedApi {  
  String imageUrl = "http://posts.doyatama.com";  
  String baseUrl = "http://posts.doyatama.com/api/";  
  
  Map<String, String> getToken() {  
    final box = GetStorage();
```

```
String? token = box.read("token");
if (token != null)
  return {
    "Authorization": "Bearer " + token,
  };
return {
  "Authorization": "Bearer " + "BadToken",
};
}
```

Pada kode tersebut berisi URL dari API yang ada serta berisi fungsi untuk mengambil token / kunci API yang akan digunakan sebagai kunci saat mengambil data dari API.

8. Buat file dengan nama `app_color.dart` pada folder `utils` yang akan digunakan untuk menyimpan warna-warna yang akan digunakan pada aplikasi:

```
class AppColor {
  static LinearGradient primaryGradient = LinearGradient(
    colors: [primary, primarySoft],
    begin: Alignment.topCenter,
    end: Alignment.bottomCenter,
  );
  static Color primary = Color(0xFF2167E6);
  static Color primarySoft = Color(0xFF184AA6);
  static Color primaryExtraSoft = Color(0xFFE9F3FC);
  static Color secondary = Color(0xFF1B1F24);
  static Color secondarySoft = Color(0xFF9D9D9D);
  static Color secondaryExtraSoft = Color(0xFFE9E9E9);
  static Color error = Color(0xFFD00E0E);
  static Color success = Color(0xFF16AE26);
  static Color warning = Color(0xFFEB8600);
}
```

9. Sebelum membuat service perlu diketahui bahwa ketika mendapatkan data dari API harus di mapping ke dalam sebuah model, sehingga data yang berasal dari API bisa dikenali dengan mudah oleh flutter. Untuk itu, silahkan buat file pada folder `lib/app/data` dengan nama `post_model.dart` untuk mapping data post, dan juga `user_model.dart` untuk maaping data pengguna.

`post_model.dart`

```
import 'dart:convert';

class PostModel {
```

```

final int? id;
final String? title;
final String? content;
final String? slug;
final int? status;

PostModel({
  this.id,
  this.title,
  this.content,
  this.slug,
  this.status,
});

factory PostModel.fromJson(Map<String, dynamic> jsonData) {
  return PostModel(
    id: jsonData['id'],
    title: jsonData['title'],
    content: jsonData['content'],
    slug: jsonData['slug'],
    status: jsonData['status_code'],
  );
}

class PostListModel {
  final int? status; // 200 - 204 - 400 - 404
  final List<PostModel>? items;

  PostListModel({this.status, this.items});

  factory PostListModel.fromJson(Map<String, dynamic> jsonData) {
    return PostListModel(
      status: jsonData['status'],
      items: jsonData['items']
        .map<PostModel>((data) => PostModel.fromJson(data))
        .toList());
  }
}

```

user_model.dart

```

class UserModel {
  final int? status;
  final int? id;
  final String? name;
  final String? email;
  final String? accessToken;
}

```

```

final String? tokenType;

UserModel({
  this.status,
  this.id,
  this.name,
  this.email,
  this.accessToken,
  this.tokenType,
});

factory UserModel.fromJson(Map<String, dynamic> jsonData) {
  return UserModel(
    status: jsonData['status'],
    id: jsonData['id'],
    name: jsonData['name'],
    email: jsonData['email'],
    accessToken: jsonData['access_token'],
    tokenType: jsonData['token_type'],
  );
}
}

```

10. Selanjutnya, perlu dibuatkan beberapa service supaya bisa melakukan komunikasi dengan API. Misalkan melakukan login, register dan CRUD post. Untuk itu buatlah file dengan nama `auth_api.dart` dan `post_api.dart` pada folder `services`.

`auth_api.dart`

File ini digunakan ketika mau login, check token, dan register. Sehingga data yang digunakan akan menyesuaikan sesuai dengan dokumentasi postman yang sudah diberikan sebelumnya.

```

import 'package:http/http.dart' as http;

class AuthApi extends SharedApi {
  // Login API
  Future<UserModel?> loginAPI(String email, String password) async {
    try {
      var jsonData;
      showLoading();
      var data = await http.post(
        Uri.parse(baseUrl + 'login'),
        body: {'email': email, 'password': password},
      );
      stopLoading();
      jsonData = json.decode(data.body);

      if (data.statusCode == 200) {

```



```

        jsonData['data']['status'] = 200;
        jsonData['data']['access_token'] = jsonData['access_token'];
        jsonData['data']['token_type'] = jsonData['token_type'];
        return UserModel.fromJson(jsonData['data']);
    } else {
        showErrorMessage(jsonData['message']);
        return UserModel.fromJson({"status": data.statusCode});
    }
} on Exception catch (_) {
    stopLoading();
    showInternetMessage("Periksa koneksi internet anda");
    return UserModel.fromJson({"status": 404});
}
}

// Check Token API
Future<UserModel?> checkTokenApi(String token) async {
    try {
        var headers = {
            "Authorization": "Bearer " + token,
        };
        var jsonData;
        showLoading();
        var data = await http.get(Uri.parse(baseUrl + 'user'), headers: headers);
        stopLoading();
        jsonData = json.decode(data.body);

        if (data.statusCode == 200) {
            jsonData['status'] = 200;
            jsonData['access_token'] = token;
            jsonData['token_type'] = "bearer";
            return UserModel.fromJson(jsonData);
        } else if (data.statusCode == 401) {
            showErrorMessage(jsonData['message']);
            return UserModel.fromJson({"status": data.statusCode});
        } else {
            showErrorMessage("Ada yang salah");
            return UserModel.fromJson({"status": data.statusCode});
        }
    } on Exception catch (_) {
        stopLoading();
        showInternetMessage("Periksa koneksi internet anda");
        return UserModel.fromJson({"status": 404});
    }
}

// Login API
Future<UserModel?> registerAPI(
    String name, String email, String password) async {
    try {

```

```

var jsonData;
showLoading();
var data = await http.post(
  Uri.parse(baseUrl + 'register'),
  body: {'name': name, 'email': email, 'password': password},
);
stopLoading();
jsonData = json.decode(data.body);

if (data.statusCode == 200) {
  jsonData['data']['status'] = 200;
  jsonData['data']['access_token'] = jsonData['access_token'];
  jsonData['data']['token_type'] = jsonData['token_type'];
  return UserModel.fromJson(jsonData['data']);
} else {
  showErrorMessage(jsonData['message']);
  return UserModel.fromJson({'status': data.statusCode});
}
} on Exception catch (_) {
  stopLoading();
  showInternetMessage("Periksa koneksi internet anda");
  return UserModel.fromJson({'status': 404});
}
}
}

```

post_api.dart

File ini digunakan ketika mau CRUD Post. Sehingga data yang digunakan akan menyesuaikan sesuai dengan dokumentasi postman yang sudah diberikan sebelumnya.

```

import 'package:http/http.dart' as http;

class PostApi extends SharedApi {
  // Login API
  Future<PostListModel> loadPostAPI() async {
    try {
      var data =
        await http.get(Uri.parse(baseUrl + 'posts'), headers: getToken());
      print("hasil" + data.statusCode.toString());

      if (data.statusCode == 200) {
        var jsonData = json.decode(data.body);
        return PostListModel.fromJson(
          {'status': 200, "items": jsonData['data']});
      } else {
        return PostListModel.fromJson({'status': data.statusCode, "items": []});
      }
    } on Exception catch (_) {

```

```

        return PostListModel.fromJson({"status": 404, "items": []});
    }
}

Future<PostModel?> addPostAPI(String title, String content) async {
  try {
    var jsonData;
    showLoading();
    var data = await http.post(
      Uri.parse(baseUrl + 'posts'),
      headers: getToken(),
      body: {'title': title, 'content': content, 'status': "1"},
    );
    stopLoading();
    jsonData = json.decode(data.body);

    if (data.statusCode == 200) {
      jsonData['data']['status_code'] = 200;
      return PostModel.fromJson(jsonData['data']);
    } else {
      showErrorMessage(jsonData['message']);
      return PostModel.fromJson({"status": data.statusCode});
    }
  } on Exception catch (_) {
    stopLoading();
    showInternetMessage("Periksa koneksi internet anda");
    return PostModel.fromJson({"status": 404});
  }
}

Future<PostModel?> editPostAPI(
  String title, String content, String id) async {
  try {
    var jsonData;
    showLoading();
    var data = await http.put(
      Uri.parse(baseUrl + 'posts/' + id.toString()),
      headers: getToken(),
      body: {'title': title, 'content': content, 'status': "1"},
    );
    stopLoading();
    jsonData = json.decode(data.body);

    if (data.statusCode == 200) {
      jsonData['data']['status_code'] = 200;
      return PostModel.fromJson(jsonData['data']);
    } else {
      showErrorMessage(jsonData['message']);
      return PostModel.fromJson({"status": data.statusCode});
    }
  }
}

```

```

    }
  } on Exception catch (_) {
    stopLoading();
    showInternetMessage("Periksa koneksi internet anda");
    return PostModel.fromJson({"status": 404});
  }
}

Future<PostModel?> deletePostAPI(String id) async {
  try {
    var jsonData;
    showLoading();
    var data = await http.delete(
      Uri.parse(baseUrl + 'posts/' + id.toString()),
      headers: getToken(),
    );
    stopLoading();
    jsonData = json.decode(data.body);

    if (data.statusCode == 200) {
      // Simpan nilai jsonData['data'] dalam variabel baru
      var postData = <String, dynamic>{};
      postData["status_code"] = 200;
      postData["status"] = 1;
      postData["id"] = 0;
      postData["title"] = "";
      postData["content"] = "";
      postData["slug"] = "";

      print(postData);

      // Kirim variabel postData ke dalam fungsi PostModel.fromJson
      return PostModel.fromJson(postData);
    } else {
      showErrorMessage(jsonData['message']);
      return PostModel.fromJson({"status": data.statusCode});
    }
  } on Exception catch (_) {
    stopLoading();
    showInternetMessage("Periksa koneksi internet anda");
    return PostModel.fromJson({"status": 404});
  }
}
}
}

```

11. Buatkan beberapa file pada folder widgets seperti berikut ini
 - message/errorMessage.dart untuk menampilkan pesan error

```

showErrorMessage(String message) {
  BotToast.showCustomText(
    align: Alignment.center,
    toastBuilder: (close) {
      return Directionality(
        textDirection: TextDirection.rtl,
        child: Container(
          color: Colors.black.withOpacity(0.45),
          alignment: Alignment.center,
          child: Container(
            width: 300,
            margin: EdgeInsets.all(10),
            decoration: BoxDecoration(
              color: Colors.white,
              borderRadius: BorderRadius.circular(8.0),
              boxShadow: [
                BoxShadow(
                  color: Colors.grey.withOpacity(0.5),
                  spreadRadius: 5,
                  blurRadius: 7,
                  offset: Offset(0, 3), // changes position of shadow
                ),
              ],
            ),
          ),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.min,
          children: [
            // Orange Bar
            Container(
              height: 70,
              width: double.infinity,
              decoration: BoxDecoration(
                color: Color(0xffE84849),
                borderRadius: BorderRadius.only(
                  topLeft: Radius.circular(8.0),
                  topRight: Radius.circular(8.0)),
              ),
              alignment: Alignment.center,
              padding: EdgeInsets.all(6),
              child: Image.asset('assets/images/error.png'),
            ),
            // Title
            Container(
              height: 56,
              alignment: Alignment.center,
              child: Text(
                "Gagal",
                style: TextStyle(
                  fontFamily: 'Droid',
                  fontSize: 24,

```



```

showInternetMessage(String message) {
  BotToast.showCustomText(
    align: Alignment.center,
    toastBuilder: (close) {
      return Directionality(
        textDirection: TextDirection.rtl,
        child: Container(
          color: Colors.black.withOpacity(0.45),
          alignment: Alignment.center,
          child: Container(
            width: 300,
            margin: EdgeInsets.all(10),
            decoration: BoxDecoration(
              color: Colors.white,
              borderRadius: BorderRadius.circular(8.0),
              boxShadow: [
                BoxShadow(
                  color: Colors.grey.withOpacity(0.5),
                  spreadRadius: 5,
                  blurRadius: 7,
                  offset: Offset(0, 3), // changes position of shadow
                ),
              ],
            ),
            child: Column(
              mainAxisAlignment: MainAxisAlignment.min,
              children: [
                // Orange Bar
                Container(
                  height: 70,
                  width: double.infinity,
                  decoration: BoxDecoration(
                    color: Color(0xFFCCA3D),
                    borderRadius: BorderRadius.only(
                      topLeft: Radius.circular(8.0),
                      topRight: Radius.circular(8.0)),
                  ),
                  alignment: Alignment.center,
                  padding: EdgeInsets.all(6),
                  child: Image.asset('assets/images/internet.png'),
                ),
                // Title
                Container(
                  height: 56,
                  alignment: Alignment.center,
                  child: Text(
                    "Gagal",
                    style: TextStyle(
                      fontFamily: 'Droid',

```


- message/successMessage.dart untuk menampilkan pesan berhasil

```
showSuccessMessage(String message) {
  BotToast.showCustomText(
    align: Alignment.center,
    toastBuilder: (close) {
      return Directionality(
        textDirection: TextDirection.rtl,
        child: Container(
          color: Colors.black.withOpacity(0.45),
          alignment: Alignment.center,
          child: Container(
            width: 300,
            margin: EdgeInsets.all(10),
            decoration: BoxDecoration(
              color: Colors.white,
              borderRadius: BorderRadius.circular(8.0),
              boxShadow: [
                BoxShadow(
                  color: Colors.grey.withOpacity(0.5),
                  spreadRadius: 5,
                  blurRadius: 7,
                  offset: Offset(0, 3), // changes position of shadow
                ),
              ],
            ),
            child: Column(
              mainAxisAlignment: MainAxisAlignment.min,
              children: [
                // Green Bar
                Container(
                  height: 70,
                  width: double.infinity,
                  decoration: BoxDecoration(
                    color: Color(0xff90b06e),
                    borderRadius: BorderRadius.only(
                      topLeft: Radius.circular(8.0),
                      topRight: Radius.circular(8.0)),
                  ),
                  alignment: Alignment.center,
                  padding: EdgeInsets.all(6),
                  child: Image.asset('assets/images/success.png'),
                ),
                // Title
                Container(
                  height: 56,
                  alignment: Alignment.center,
                  child: Text(
                    "Berhasil",
                    style: TextStyle(
```


- auto_load.dart digunakan untuk memanggil fungsi saat pertama kali

```
class AutoLoad extends StatefulWidget {
  final Function? onInit;
  final Widget? child;
  const AutoLoad({@required this.onInit, @required this.child});
  @override
  _AutoLoadState createState() => _AutoLoadState();
}

class _AutoLoadState extends State<AutoLoad> {
  @override
  void initState() {
    super.initState();
    WidgetsBinding.instance?.addPostFrameCallback((_) {
      print("WidgetsBinding");

      if (widget.onInit != null) {
        widget.onInit!();
      }
    });
  }

  void runFunction() {
    if (widget.onInit != null) {
      widget.onInit!();
    }
    // super.initState();
  }

  @override
  Widget build(BuildContext context) {
    return Directionality(
      textDirection: TextDirection.ltr,
      child: widget.child!,
    );
  }
}
```

- custom_alert_dialog.dart merupakan widget bawaan Flutter yang digunakan untuk menampilkan jendela notifikasi atau peringatan kepada pengguna.

```
class CustomAlertDialog {
  static confirmAdmin({
    required String title,
```

```

required String message,
required void Function() onConfirm,
required void Function() onCancel,
required TextEditingController controller,
}) {
  Get.defaultDialog(
    title: "",
    contentPadding: EdgeInsets.symmetric(horizontal: 24, vertical: 0),
    radius: 8,
    titlePadding: EdgeInsets.zero,
    titleStyle: TextStyle(fontSize: 0),
    content: Column(
      children: [
        Container(
          margin: EdgeInsets.symmetric(vertical: 24),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Text(
                title,
                style: TextStyle(
                  fontSize: 16,
                  fontFamily: 'poppins',
                  fontWeight: FontWeight.w600,
                ),
              ),
              SizedBox(height: 16),
              Text(
                message,
                style: TextStyle(
                  color: AppColor.secondarySoft,
                  height: 150 / 100,
                ),
              ),
            ],
          ),
        ),
        CustomInput(
          margin: EdgeInsets.only(bottom: 24),
          controller: controller,
          label: 'password',
          hint: '*****',
          obscureText: true,
        ),
        Container(
          margin: EdgeInsets.only(bottom: 16),
          child: Row(
            children: [
              Expanded(
                flex: 6,

```

```

        child: ElevatedButton(
          onPressed: onCancel,
          child: Text(
            "batal",
            style: TextStyle(color: AppColor.secondarySoft),
          ),
          style: ElevatedButton.styleFrom(
            padding: EdgeInsets.symmetric(vertical: 12),
            primary: AppColor.primaryExtraSoft,
            elevation: 0,
            onPressed: AppColor.primary,
          ),
        ),
      ),
    ),
    SizedBox(width: 16),
    Expanded(
      flex: 6,
      child: ElevatedButton(
        onPressed: onConfirm,
        child: Text("Iya"),
        style: ElevatedButton.styleFrom(
          primary: AppColor.primary,
          padding: EdgeInsets.symmetric(vertical: 12),
          elevation: 0,
        ),
      ),
    ),
  ],
),
],
),
),
);
}

static showPresenceAlert({
  required String title,
  required String message,
  required void Function() onConfirm,
  required void Function() onCancel,
}) {
  Get.defaultDialog(
    title: "",
    contentPadding: EdgeInsets.symmetric(horizontal: 24, vertical: 0),
    radius: 8,
    titlePadding: EdgeInsets.zero,
    titleStyle: TextStyle(fontSize: 0),
    content: Column(
      children: [
        Container(

```

```

margin: EdgeInsets.only(bottom: 32, top: 24),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.start,
  children: [
    Text(
      title,
      style: TextStyle(
        fontSize: 16,
        fontFamily: 'poppins',
        fontWeight: FontWeight.w600,
      ),
    ),
    SizedBox(height: 16),
    Text(
      message,
      style: TextStyle(
        color: AppColor.secondarySoft,
        height: 150 / 100,
      ),
    ),
  ],
),
),
Container(
  margin: EdgeInsets.only(bottom: 16),
  child: Row(
    children: [
      Expanded(
        flex: 6,
        child: ElevatedButton(
          onPressed: onCancel,
          child: Text(
            "Batal",
            style: TextStyle(color: AppColor.secondarySoft),
          ),
          style: ElevatedButton.styleFrom(
            padding: EdgeInsets.symmetric(vertical: 12),
            primary: AppColor.primaryExtraSoft,
            elevation: 0,
            onPressed: AppColor.primary,
          ),
        ),
      ),
      Expanded(
        flex: 6,
        child: ElevatedButton(
          onPressed: onConfirm,
          child: Text("Iya"),
          style: ElevatedButton.styleFrom(

```

```

        primary: AppColor.primary,
        padding: EdgeInsets.symmetric(vertical: 12),
        elevation: 0,
      ),
    ),
  ),
  ],
),
),
  ],
),
);
}
}

```

- custom_input.dart digunakan untuk memberikan style pada input sehingga lebih bagus untuk UI nya.

```

class CustomInput extends StatefulWidget {
  final TextEditingController controller;
  final String label;
  final String hint;
  final bool disabled;
  final EdgeInsetsGeometry margin;
  final bool obscureText;
  final Widget? suffixIcon;
  final bool isDate;
  final bool isNumber;
  final bool isClickEmpty;
  CustomInput({
    required this.controller,
    required this.label,
    required this.hint,
    this.disabled = false,
    this.margin = const EdgeInsets.only(bottom: 16),
    this.obscureText = false,
    this.isDate = false,
    this.suffixIcon,
    this.isNumber = false,
    this.isClickEmpty = false,
  });

  @override
  State<CustomInput> createState() => _CustomInputState();
}

class _CustomInputState extends State<CustomInput> {

```

```

@override
Widget build(BuildContext context) {
  print("buildded");
  return Material(
    color: Colors.white,
    child: Container(
      width: MediaQuery.of(context).size.width,
      padding: EdgeInsets.only(left: 14, right: 14, top: 4),
      margin: widget.margin,
      decoration: BoxDecoration(
        color: (widget.disabled == false)
          ? Colors.transparent
          : AppColor.primaryExtraSoft,
        borderRadius: BorderRadius.circular(8),
        border: Border.all(width: 1, color: AppColor.secondaryExtraSoft),
      ),
      child: TextField(
        readOnly: widget.disabled,
        obscureText: widget.obscureText,
        style: TextStyle(fontSize: 14, fontFamily: 'poppins'),
        maxLines: 1,
        controller: widget.controller,
        keyboardType:
          widget.isNumber ? TextInputType.number : TextInputType.text,
        onTap: () async {
          if (widget.isDate) {
            DateTime? pickedDate = await showDatePicker(
              context: context,
              initialDate: DateTime.now(),
              firstDate: DateTime(
                1900, //DateTime.now() - not to allow to choose before today.
                lastDate: DateTime.now(),
              );
            if (pickedDate != null) {
              print(
                pickedDate); //pickedDate output format => 2021-03-10 00:00:00.000
              String formattedDate =
                DateFormat('dd-MM-yyyy').format(pickedDate);
              print(
                formattedDate); //formatted date output using intl package => 2021-03-16
              //you can implement different kind of Date Format here according to your
requirement

              setState() {
                widget.controller.text =
                  formattedDate; //set output date to TextField value.
              };
            } else {
              print("Date is not selected");
            }
          }
        }
      )
    )
  );
}

```



```

    }
    if (widget.isClickEmpty) {
      setState() {
        widget.controller.text = "";
      };
    }
  },
  decoration: InputDecoration(
    suffixIcon: widget.suffixIcon ?? SizedBox(),
    label: Text(
      widget.label,
      style: TextStyle(
        color: AppColor.secondarySoft,
        fontSize: 14,
      ),
    ),
    floatingLabelBehavior: FloatingLabelBehavior.always,
    border: InputBorder.none,
    hintText: widget.hint,
    hintStyle: TextStyle(
      fontSize: 14,
      fontFamily: 'poppins',
      fontWeight: FontWeight.w500,
      color: AppColor.secondarySoft,
    ),
  ),
),
),
),
),
);
}
}

```

- loading.dart digunakan untuk membuat UI loading sehingga bisa mengetahui bahwa sedang ada proses yang berjalan

```

class LoadingWidget extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Container(
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(8),
      ),
      width: 180,
      height: 100,
      padding: EdgeInsets.all(0),
      child: Image.asset('assets/gif/Loading.gif'),
    );
  }
}

```

```

    }
  }

  showLoading() {
    BotToast.showCustomLoading(toastBuilder: (close) {
      return LoadingWidget();
    });
  }

  stopLoading() {
    BotToast.closeAllLoading();
  }

```

- no_data.dart digunakan untuk memberikan kondisi ketika tidak ada data yang ditampilkan adalah no_data ini

```

class NoData extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Container(
      color: Colors.white,
      width: double.infinity,
      height: double.infinity,
      alignment: Alignment.center,
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Image.asset(
            'assets/img/empty.png',
            height: 180,
            width: 180,
          ),
          SizedBox(
            height: 20,
          ),
          Text(
            "Data Kosong",
            style: GoogleFonts.tajawal(
              fontSize: 24,
              fontWeight: FontWeight.w500,
              color: Color(0xffffca54)),
          ),
          SizedBox(
            height: 20,
          ),
          Container(
            margin: EdgeInsets.fromLTRB(20, 0, 20, 0),

```

```

        child: Text(
          "Data dari server masih belum ada nih ...",
          style: GoogleFonts.cairo(
            color: Color(0xff777777),
          ),
          textAlign: TextAlign.center,
        ),
      ),
    ),
    SizedBox(
      height: 30,
    ),
  ],
),
);
}
}

```

- `no_network.dart` digunakan untuk menampilkan halaman bahwa ketika tidak ada internet.

```

class NoNetwork extends StatelessWidget {
  final Function? onInit;
  const NoNetwork({Key? key, @required this.onInit}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return Container(
      color: Colors.white,
      width: double.infinity,
      height: double.infinity,
      alignment: Alignment.center,
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Image.asset(
            'assets/images/no-internet.png',
            height: 180,
            width: 180,
          ),
          SizedBox(
            height: 20,
          ),
          Text(
            "Terputus",
            style: GoogleFonts.tajawal(
              fontSize: 24,
              fontWeight: FontWeight.w500,
            ),
          ),
        ],
      ),
    );
  }
}

```

```

        color: Color(0xffffca54)),
    ),
    SizedBox(
      height: 20,
    ),
    Container(
      margin: EdgeInsets.fromLTRB(20, 0, 20, 0),
      child: Text(
        "Maaf, koneksi internet tidak tersedia, periksa koneksi internet Anda dan coba lagi",
        style: GoogleFonts.cairo(
          color: Color(0xff777777),
        ),
        textAlign: TextAlign.center,
      ),
    ),
    SizedBox(
      height: 30,
    ),
    Container(
      height: 52,
      width: 160,
      child: ElevatedButton(
        onPressed: () async {
          if (onInit != null) {
            await onInit!();
          }
        },
        child: Text(
          "Muat Ulang",
          style: GoogleFonts.cairo(
            fontSize: 16,
          ),
        ),
        style: ElevatedButton.styleFrom(
          primary: Color(0xffffca54),
        ),
      ),
    ),
  ],
),
);
}
}

```

12. Membuat halaman Splash Screen

- Ketikkan perintah berikut pada terminal untuk membuat halaman splash pada Get X:

get create page:start

- Atur file start_controller.dart menjadi seperti berikut ini :
Deklarasi variabel untuk model dan juga penyimpanan Get Storage

```
UserModel? userModel;  
final box = GetStorage();  
bool startScreen = false;
```

Buat fungsi autoLogin yang digunakan untuk melakukan pengecekan saat halaman masih ada pada splash screen.

```
autoLogin() async {  
  startScreen = false;  
  update();  
  if (box.hasData("token") == true) {  
    userModel = await AuthApi().checkTokenApi(box.read("token"));  
    if (userModel!.status == 200) {  
      Get.offAndToNamed(Routes.HOME);  
    } else if (userModel!.status == 404) {  
      startScreen = true;  
      update();  
    } else if (userModel!.status == 401) {  
      box.remove("token");  
      box.remove("id");  
      Get.offAndToNamed(Routes.LOGIN);  
    } else {  
      box.remove("token");  
      box.remove("id");  
      Get.offAndToNamed(Routes.LOGIN);  
    }  
  } else {  
    Get.offAndToNamed(Routes.LOGIN);  
  }  
}
```

Berikut kode lengkap untuk start_controller.dart

```
class StartController extends GetxController {  
  UserModel? userModel;  
  final box = GetStorage();  
  bool startScreen = false;  
  
  autoLogin() async {  
    startScreen = false;
```

```

update();
if (box.hasData("token") == true) {
  userModel = await AuthApi().checkTokenApi(box.read("token"));
  if (userModel!.status == 200) {
    Get.offAndToNamed(Routes.HOME);
  } else if (userModel!.status == 404) {
    startScreen = true;
    update();
  } else if (userModel!.status == 401) {
    box.remove("token");
    box.remove("id");
    Get.offAndToNamed(Routes.LOGIN);
  } else {
    box.remove("token");
    box.remove("id");
    Get.offAndToNamed(Routes.LOGIN);
  }
} else {
  Get.offAndToNamed(Routes.LOGIN);
}
}
}
}

```

- Atur file start_view.dart menjadi seperti berikut ini :

```

class StartView extends GetView<StartController> {
  const StartView({Key? key}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    return GetBuilder<StartController>(
      builder: (controller) => AutoLoad(
        onInit: () async {
          await new Future.delayed(const Duration(seconds: 3));
          await controller.autoLogin();
        },
        child: Scaffold(
          body: GetBuilder<StartController>(
            builder: (controller) => controller.startScreen == false
              ? Container(
                  color: AppColor.primary,
                  width: double.infinity,
                  height: double.infinity,
                  alignment: Alignment.center,
                  child: Column(
                    mainAxisAlignment: MainAxisAlignment.center,
                    children: [
                      Spacer(),
                      Icon(Icons.account_tree_outlined,

```

```

        size: 48, color: Colors.white),
      SizedBox(
        height: 20,
      ),
      Text(
        "Post app",
        style: GoogleFonts.tajawal(
          fontSize: 24,
          fontWeight: FontWeight.w500,
          color: Color(0xffffca54)),
      ),
      SizedBox(
        height: 20,
      ),
      Container(
        margin: EdgeInsets.fromLTRB(20, 0, 20, 0),
        child: Column(
          children: [
            Text(
              "Belajar Bersama",
              style: GoogleFonts.cairo(
                color: Colors.white,
              ),
              textAlign: TextAlign.center,
            ),
            Text(
              "Building the Nation Through Education",
              style: GoogleFonts.cairo(
                color: Colors.white,
              ),
              textAlign: TextAlign.center,
            ),
          ],
        )),
      Spacer(),
      Text(
        "Hummasoft. V1",
        style: GoogleFonts.cairo(
          color: Colors.white,
          fontSize: 12,
        ),
        textAlign: TextAlign.center,
      ),
      SizedBox(
        height: 30,
      ),
    ],
  ),
)
: NoNetwork(

```

```

        onInit: controller.autoLogin,
      ),
    ),
  );
}
}

```

- Atur initial route pada file lib/app/routes/app_pages.dart menjadi seperti berikut ini untuk menjadikan halaman start_view menjadi halaman yang pertama kali dijalankan.

```
static const INITIAL = Routes.START;
```

- Selesai, silahkan di jalankan dengan menggunakan perintah `flutter run`.

13. Membuat halaman Login

- Ketikkan perintah berikut pada terminal untuk membuat halaman login pada Get X:

```
get create page:login
```

- Atur file login_controller.dart menjadi seperti berikut ini :

Deklarasi variabel untuk model pengguna dan juga untuk widget input

```

UserModel? userModel;
final box = GetStorage();

bool loginScreen = false;
RxBool isLoading = false.obs;
RxBool obscureText = true.obs;
TextEditingController emailC = TextEditingController();
TextEditingController passC = TextEditingController();

```

Buatkan fungsi untuk login

```

Future login() async {
  loginScreen = false;
  update();
}

```



```

userModel = await AuthApi().loginAPI(emailC.text, passC.text);
if (userModel!.status == 200) {
  await box.write("token", userModel!.accessToken);
  await box.write("name", userModel!.name);
  await box.write("email", userModel!.email);
  update();
  Get.offAndToNamed(Routes.HOME);
} else if (userModel!.status == 404) {
  loginScreen = true;
  update();
}
}

```

Berikut kode lengkap untuk login_controller.dart

```

class LoginController extends GetxController {
  UserModel? userModel;
  final box = GetStorage();

  bool loginScreen = false;
  RxBool isLoading = false.obs;
  RxBool obscureText = true.obs;
  TextEditingController emailC = TextEditingController();
  TextEditingController passC = TextEditingController();

  Future login() async {
    loginScreen = false;
    update();
    userModel = await AuthApi().loginAPI(emailC.text, passC.text);
    if (userModel!.status == 200) {
      await box.write("token", userModel!.accessToken);
      await box.write("name", userModel!.name);
      await box.write("email", userModel!.email);
      update();
      Get.offAndToNamed(Routes.HOME);
    } else if (userModel!.status == 404) {
      loginScreen = true;
      update();
    }
  }
}

```

- Atur file login_view.dart menjadi seperti berikut ini :

```

class LoginView extends GetView<LoginController> {
  const LoginView({Key? key}) : super(key: key);
  @override

```

```

Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      backgroundColor: AppColor.primary,
      elevation: 0.0,
      titleSpacing: 10.0,
      centerTitle: true,
      title: Text("Login"),
    ),
    backgroundColor: AppColor.primary,
    body: ListView(
      shrinkWrap: true,
      children: [
        Container(
          height: MediaQuery.of(context).size.height * 25 / 100,
          width: MediaQuery.of(context).size.width,
          padding: EdgeInsets.only(left: 32),
          decoration: BoxDecoration(
            gradient: AppColor.primaryGradient,
            image: DecorationImage(
              image: AssetImage('assets/images/pattern-1-1.png'),
              fit: BoxFit.cover,
            ),
          ),
        ),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text(
              "POSTS App",
              style: TextStyle(
                fontSize: 24,
                color: Colors.white,
                fontFamily: 'poppins',
                height: 150 / 100,
                fontWeight: FontWeight.w600,
              ),
            ),
            SizedBox(height: 10),
            Text(
              "by Hummasoft",
              style: TextStyle(color: Colors.white),
            ),
          ],
        ),
        Container(
          height: MediaQuery.of(context).size.height * 65 / 100,
          width: MediaQuery.of(context).size.width,
          color: Colors.white,

```

```

padding: EdgeInsets.only(left: 20, right: 20, top: 36, bottom: 84),
child: Column(
  mainAxisAlignment: MainAxisAlignment.start,
  crossAxisAlignment: CrossAxisAlignment.start,
  children: [
    Container(
      margin: EdgeInsets.only(bottom: 24),
      child: Text(
        'Masuk',
        style: TextStyle(
          fontSize: 18,
          fontFamily: 'poppins',
          fontWeight: FontWeight.w600,
        ),
      ),
    ),
    Container(
      width: MediaQuery.of(context).size.width,
      padding: EdgeInsets.only(left: 14, right: 14, top: 4),
      margin: EdgeInsets.only(bottom: 16),
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(8),
        border: Border.all(
          width: 1, color: AppColor.secondaryExtraSoft),
      ),
      child: TextField(
        style: TextStyle(fontSize: 14, fontFamily: 'poppins'),
        maxLines: 1,
        controller: controller.emailC,
        keyboardType: TextInputType.emailAddress,
        decoration: InputDecoration(
          label: Text(
            "Email",
            style: TextStyle(
              color: AppColor.secondarySoft,
              fontSize: 14,
            ),
          ),
          floatingLabelBehavior: FloatingLabelBehavior.always,
          border: InputBorder.none,
          hintText: "email@gmail.com",
          hintStyle: TextStyle(
            fontSize: 14,
            fontFamily: 'poppins',
            fontWeight: FontWeight.w500,
            color: AppColor.secondarySoft,
          ),
        ),
      ),
    ),
  ],
),

```



```

        side: BorderSide(
          color: AppColor
            .primary, // Replace this with your desired border color
          width:
            1.0, // Replace this with your desired border width
        ),
      ),
    ),
  ),
),
1,
),
),
1,
),
),
);
}
}

```

- Selesai, silahkan di jalankan dengan menggunakan perintah `flutter run`.

14. Membuat halaman Register

- Ketikkan perintah berikut pada terminal untuk membuat halaman register pada Get X:

```
get create page:register
```

- Atur file register_controller.dart menjadi seperti berikut ini :

Deklarasi variabel untuk model pengguna dan juga untuk widget input

```

UserModel? userModel;
final box = GetStorage();

RxBool isLoading = false.obs;
RxBool obscureText = true.obs;
TextEditingController nameC = TextEditingController();
TextEditingController emailC = TextEditingController();
TextEditingController passC = TextEditingController();

```

Buatkan fungsi untuk registrasi yang akan menggunakan service auth_service.dart untuk pemanggilan API nya:

```

Future registration() async {
  update();
  userModel =
    await AuthApi().registerAPI(nameC.text, emailC.text, passC.text);
  if (userModel!.status == 200) {
    await box.write("token", userModel!.accessToken);
    await box.write("name", userModel!.name);
    await box.write("email", userModel!.email);
    update();
    Get.offAndToNamed(Routes.HOME);
  } else if (userModel!.status == 404) {
    update();
  }
}

```

Berikut kode lengkap untuk register_controller.dart

```

class RegisterController extends GetxController {
  UserModel? userModel;
  final box = GetStorage();

  RxBool isLoading = false.obs;
  RxBool obscureText = true.obs;
  TextEditingController nameC = TextEditingController();
  TextEditingController emailC = TextEditingController();
  TextEditingController passC = TextEditingController();

  Future registration() async {
    update();
    userModel =
      await AuthApi().registerAPI(nameC.text, emailC.text, passC.text);
    if (userModel!.status == 200) {
      await box.write("token", userModel!.accessToken);
      await box.write("name", userModel!.name);
      await box.write("email", userModel!.email);
      update();
      Get.offAndToNamed(Routes.HOME);
    } else if (userModel!.status == 404) {
      update();
    }
  }
}

```

- Atur file register_view.dart menjadi seperti berikut ini :

```

class RegisterView extends GetView<RegisterController> {
  const RegisterView({Key? key}) : super(key: key);
}

```

```

@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: AppColor.primary,
    appBar: AppBar(
      automaticallyImplyLeading: true,
      elevation: 0.0,
      titleSpacing: 10.0,
      centerTitle: true,
      title: Text("Pendaftaran"),
      leading: IconButton(
        icon: SvgPicture.asset(
          'assets/icons/arrow-left.svg',
          color: Colors.white,
        ),
        onPressed: () => Get.back(),
      ),
      backgroundColor: Colors.transparent,
    ),
    extendBodyBehindAppBar: true,
    body: ListView(
      shrinkWrap: true,
      children: [
        Container(
          height: MediaQuery.of(context).size.height * 25 / 100,
          width: MediaQuery.of(context).size.width,
          padding: EdgeInsets.only(left: 32),
          decoration: BoxDecoration(
            gradient: AppColor.primaryGradient,
            image: DecorationImage(
              image: AssetImage('assets/images/pattern-1-1.png'),
              fit: BoxFit.cover,
            ),
          ),
        ),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text(
              "POSTS App",
              style: TextStyle(
                fontSize: 24,
                color: Colors.white,
                fontFamily: 'poppins',
                height: 150 / 100,
                fontWeight: FontWeight.w600,
              ),
            ),
            SizedBox(height: 10),

```



```

        Text(
          "by Hummasoft",
          style: TextStyle(color: Colors.white),
        ),
      ],
    ),
  ),
  Container(
    height: MediaQuery.of(context).size.height * 70 / 100,
    width: MediaQuery.of(context).size.width,
    color: Colors.white,
    padding: EdgeInsets.only(left: 20, right: 20, top: 36, bottom: 84),
    child: Column(
      mainAxisAlignment: MainAxisAlignment.start,
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        Container(
          margin: EdgeInsets.only(bottom: 24),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Text(
                'Daftar',
                style: TextStyle(
                  fontSize: 18,
                  fontFamily: 'poppins',
                  fontWeight: FontWeight.w600,
                ),
              ),
            ],
          ),
        ),
      ],
    ),
    Container(
      width: MediaQuery.of(context).size.width,
      padding: EdgeInsets.only(left: 14, right: 14, top: 4),
      margin: EdgeInsets.only(bottom: 16),
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(8),
        border: Border.all(
          width: 1, color: AppColor.secondaryExtraSoft),
      ),
      child: TextField(
        style: TextStyle(fontSize: 14, fontFamily: 'poppins'),
        maxLines: 1,
        controller: controller.nameC,
        decoration: InputDecoration(
          label: Text(
            "Nama Lengkap",
            style: TextStyle(

```



```

Container(
  margin: EdgeInsets.only(bottom: 24),
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Material(
        child: Container(
          width: MediaQuery.of(context).size.width,
          padding: EdgeInsets.only(left: 14, right: 14, top: 4),
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(8),
            border: Border.all(
              width: 1, color: AppColor.secondaryExtraSoft),
          ),
        child: Obx(
          () => TextField(
            style: TextStyle(
              fontSize: 14, fontFamily: 'poppins'),
            maxLines: 1,
            controller: controller.passC,
            obscureText: controller.obsecureText.value,
            decoration: InputDecoration(
              label: Text(
                "Kata Sandi",
                style: TextStyle(
                  color: AppColor.secondarySoft,
                  fontSize: 14,
                ),
              ),
              floatingLabelBehavior:
                FloatingLabelBehavior.always,
              border: InputBorder.none,
              hintText: "*****",
              suffixIcon: IconButton(
                icon: (controller.obsecureText != false)
                  ? SvgPicture.asset(
                      'assets/icons/show.svg')
                  : SvgPicture.asset(
                      'assets/icons/hide.svg'),
                onPressed: () {
                  controller.obsecureText.value =
                    !(controller.obsecureText.value);
                },
              ),
              hintStyle: TextStyle(
                fontSize: 14,
                fontFamily: 'poppins',
                fontWeight: FontWeight.w500,
                color: AppColor.secondarySoft,
              ),
            ),
          ),
        ),
      ),
    ],
  ),
)

```



```
);  
}  
}
```

- Selesai, silahkan di jalankan dengan menggunakan perintah `flutter run`.

15. Mengatur halaman Home

- Atur file home_controller.dart menjadi seperti berikut ini :

Deklarasi variabel untuk model post, karena pada halaman home akan menampilkan post.

```
PostListModel? posts;  
final box = GetStorage();  
bool homeScreen = false;
```

Buatkan fungsi loadPost yang akan digunakan untuk memanggil data dari API dengan memanfaatkan service post_service.

```
loadPost() async {  
  homeScreen = false;  
  update();  
  showLoading();  
  posts = await PostApi().loadPostAPI();  
  update();  
  stopLoading();  
  if (posts?.status == 200) {  
  } else if (posts!.status == 204) {  
    print("Empty");  
  } else if (posts!.status == 404) {  
    homeScreen = true;  
    update();  
  } else if (posts!.status == 401) {  
  } else {  
    print("someting wrong 400");  
  }  
}
```

Buatkan fungsi logout yang akan digunakan sebagai trigger ketika mau keluar dari sistem

```
logout() async {  
  CustomAlertDialog.showPresenceAlert(  
    title: "Logout System",
```

```

message: "Apakah anda yakin akan logout dari sistem ?",
onCancel: () => Get.back(),
onConfirm: () async {
  Get.back(); // close modal
  Get.back(); // back page
  try {
    box.remove("token");
    box.remove("name");
    box.remove("email");
    update();
    await Get.offAllNamed(Routes.LOGIN);
  } catch (e) {}
},
);
}

```

Berikut kode lengkap untuk home_controller.dart

```

class HomeController extends GetxController {
  PostListModel? posts;
  final box = GetStorage();
  bool homeScreen = false;

  loadPost() async {
    homeScreen = false;
    update();
    showLoading();
    posts = await PostApi().loadPostAPI();
    update();
    stopLoading();
    if (posts?.status == 200) {
    } else if (posts!.status == 204) {
      print("Empty");
    } else if (posts!.status == 404) {
      homeScreen = true;
      update();
    } else if (posts!.status == 401) {
    } else {
      print("something wrong 400");
    }
  }

  logout() async {
    CustomAlertDialog.showPresenceAlert(
      title: "Logout System",
      message: "Apakah anda yakin akan logout dari sistem ?",
      onCancel: () => Get.back(),
      onConfirm: () async {

```

```

    Get.back(); // close modal
    Get.back(); // back page
    try {
      box.remove("token");
      box.remove("name");
      box.remove("email");
      update();
      await Get.offAllNamed(Routes.LOGIN);
    } catch (e) {}
  },
);
}
}

```

- Atur file `home_view.dart` menjadi seperti berikut ini :

```

class HomeView extends GetView<HomeController> {
  const HomeView({Key? key}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    return GetBuilder<HomeController>(
      builder: (controller) => AutoLoad(
        onInit: () async {
          await controller.loadPost();
        },
        child: Scaffold(
          extendBody: true,
          body: controller.homeScreen == false
            ? ListView(
                shrinkWrap: true,
                physics: BouncingScrollPhysics(),
                padding: EdgeInsets.symmetric(horizontal: 20, vertical: 36),
                children: [
                  SizedBox(height: 16),
                  // Section 1 - Welcome Back
                  Container(
                    margin: EdgeInsets.only(bottom: 16),
                    width: MediaQuery.of(context).size.width,
                    child: Row(
                      mainAxisAlignment: MainAxisAlignment.spaceBetween,
                      children: [
                        Row(
                          children: [
                            ClipOval(
                              child: Container(
                                width: 42,
                                height: 42,
                                child: Image.network(
                                  "https://ui-avatars.com/api/?name=${controller.box.read('name')}",

```

```

        fit: BoxFit.cover,
      ),
    ),
  ),
  SizedBox(width: 24),
  Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Text(
        "Selamat datang kembali",
        style: TextStyle(
          fontSize: 12,
          color: AppColor.secondarySoft,
        ),
      ),
      SizedBox(height: 4),
      Text(
        controller.box.read('name'),
        style: TextStyle(
          fontWeight: FontWeight.w500,
          fontFamily: 'poppins',
        ),
      ),
    ],
  ),
],
),
],
),
InkWell(
  onTap: () => {controller.logout()},
  child: Text("Logout"),
),
],
),
),
// section 2 - card
Container(
  width: MediaQuery.of(context).size.width,
  padding: EdgeInsets.only(
    left: 24, top: 24, right: 24, bottom: 16),
  decoration: BoxDecoration(
    color: AppColor.primary,
    borderRadius: BorderRadius.circular(8),
    image: DecorationImage(
      image: AssetImage('assets/images/pattern-1.png'),
      fit: BoxFit.cover,
    ),
  ),
),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.start,
  children: [

```



```

// job
Text(
  controller.box.read('email'),
  style: TextStyle(
    color: Colors.white,
    fontFamily: 'poppins',
    fontWeight: FontWeight.w500,
  ),
),
SizedBox(
  height: 20,
),
Container(
  width: MediaQuery.of(context).size.width,
  padding: EdgeInsets.symmetric(
    horizontal: 8,
    vertical: 16,
  ),
  decoration: BoxDecoration(
    color: AppColor.primarySoft,
    borderRadius: BorderRadius.circular(8),
  ),
  child: Row(
    children: [
      GetBuilder<HomeController>(
        builder: (controller) => controller
          .posts!.status ==
            200
        ? Expanded(
            child: Column(
              children: [
                Container(
                  margin:
                    EdgeInsets.only(bottom: 6),
                  child: Text(
                    "Jumlah Todo",
                    style: TextStyle(
                      fontSize: 12,
                      color: Colors.white,
                    ),
                  ),
                ),
              ],
            ),
          ),
      Text(
        "${controller.posts!.items!.length.toString()}",
        style: TextStyle(
          fontSize: 16,
          fontWeight: FontWeight.w700,
          color: Colors.white,
        ),
      ),
    ],
  ),
)

```



```

GetBuilder<HomeController>(
  builder: (controller) => controller.posts!.status == 200
    ? ListView.separated(
      itemCount: controller.posts!.items!.length,
      shrinkWrap: true,
      physics: BouncingScrollPhysics(),
      separatorBuilder: (context, index) =>
        SizedBox(height: 16),
      itemBuilder: (context, index) {
        var postData = controller.posts!.items![index];
        return InkWell(
          onTap: () => {
            Get.toNamed(
              Routes.DETAIL_POST,
              arguments: {
                "id": "${postData.id}",
                "title": "${postData.title}",
                "content": "${postData.content}",
              },
            ),
          },
        ),
      },
      borderRadius: BorderRadius.circular(8),
      child: Container(
        width: MediaQuery.of(context).size.width,
        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(8),
          border: Border.all(
            width: 1,
            color: AppColor.primaryExtraSoft,
          ),
        ),
        padding: EdgeInsets.only(
          left: 24,
          top: 20,
          right: 29,
          bottom: 20),
        child: Row(
          mainAxisAlignment:
            MainAxisAlignment.spaceBetween,
          children: [

```



```
get create page:all_post
```

- Atur file all_post_controller.dart menjadi seperti berikut ini :

Deklarasi variabel model post dan juga penyimpana Get Storage

```
PostListModel? posts;  
final box = GetStorage();  
bool homeScreen = false;
```

Tambahkan fungsi loadPost yang digunakan untuk memanggil data dari API dengan menggunakan service post_service

```
loadPost() async {  
  homeScreen = false;  
  update();  
  showLoading();  
  posts = await PostApi().loadPostAPI();  
  update();  
  stopLoading();  
  if (posts?.status == 200) {  
  } else if (posts!.status == 204) {  
    print("Empty");  
  } else if (posts!.status == 404) {  
    homeScreen = true;  
    update();  
  } else if (posts!.status == 401) {  
  } else {  
    print("someting wrong 400");  
  }  
}
```

Berikut kode lengkap untuk all_post_controller.dart

```
class AllPostController extends GetxController {  
  PostListModel? posts;  
  final box = GetStorage();  
  bool homeScreen = false;  
  
  loadPost() async {  
    homeScreen = false;  
    update();  
    showLoading();  
    posts = await PostApi().loadPostAPI();  
  }  
}
```

```

update();
stopLoading();
if (posts?.status == 200) {
} else if (posts!.status == 204) {
  print("Empty");
} else if (posts!.status == 404) {
  homeScreen = true;
  update();
} else if (posts!.status == 401) {
} else {
  print("something wrong 400");
}
}
}
}

```

- Atur file `all_post_view.dart` menjadi seperti berikut ini :

```

class AllPostView extends GetView<AllPostController> {
  const AllPostView({Key? key}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    return GetBuilder<AllPostController>(
      builder: (controller) => AutoLoad(
        onInit: () async {
          await controller.loadPost();
        },
        child: Scaffold(
          appBar: AppBar(
            title: Text(
              'Semua Data Post',
              style: TextStyle(
                color: AppColor.secondary,
                fontSize: 14,
              ),
            ),
            leading: IconButton(
              onPressed: () => Get.back(),
              icon: SvgPicture.asset('assets/icons/arrow-left.svg'),
            ),
            backgroundColor: Colors.white,
            elevation: 0,
            centerTitle: true,
            bottom: PreferredSize(
              preferredSize: Size.fromHeight(1),
              child: Container(
                width: MediaQuery.of(context).size.width,
                height: 1,
                color: AppColor.secondaryExtraSoft,
              ),
            ),
          ),
        ),
      ),
    );
  }
}

```

```

    ),
  ),
  body: GetBuilder<AllPostController>(
    builder: (controller) => controller.posts!.status == 200
    ? ListView.separated(
      itemCount: controller.posts!.items!.length,
      shrinkWrap: true,
      physics: BouncingScrollPhysics(),
      separatorBuilder: (context, index) => SizedBox(height: 16),
      itemBuilder: (context, index) {
        var postData = controller.posts!.items![index];
        return InkWell(
          onTap: () => {
            Get.toNamed(
              Routes.DETAIL_POST,
              arguments: {
                "id": "${postData.id}",
                "title": "${postData.title}",
                "content": "${postData.content}",
              },
            ),
          },
        ),
      },
      borderRadius: BorderRadius.circular(8),
      child: Container(
        width: MediaQuery.of(context).size.width,
        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(8),
          border: Border.all(
            width: 1,
            color: AppColor.primaryExtraSoft,
          ),
        ),
        padding: EdgeInsets.only(
          left: 24, top: 20, right: 29, bottom: 20),
        child: Row(
          mainAxisAlignment: MainAxisAlignment.spaceBetween,
          children: [
            Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text(
                  (postData.title == null)
                    ? "-"
                    : "${postData.title}",
                  style: TextStyle(fontSize: 12),
                ),
                Text(
                  "${postData.content}",
                  style: TextStyle(fontSize: 12),
                ),
              ],
            ),
          ],
        ),
      ),
    ),
  ),

```

```

        ],
      ),
    ],
  ),
);
},
)
: NoData(),
),
),
),
);
}
}
}

```

- Selesai, silahkan di jalankan dengan menggunakan perintah `flutter run`.

17. Membuat halaman Tambah Post

- Ketikkan perintah berikut pada terminal untuk membuat halaman tambah post pada Get X:

```
get create page:add_post
```

- Atur file add_post_controller.dart menjadi seperti berikut ini :

Definisi variabel model post dan juga untuk widget text nya yang akan berisi title dan juga content

```

PostModel? postModel;

RxBool isLoading = false.obs;
RxBool isLoadingCreateTodo = false.obs;

TextEditingController titleC = TextEditingController();
TextEditingController contentC = TextEditingController();

```

Tambahkan fungsi onClose, yang digunakan untuk menghilangkan controller yang sudah di definisikan.

```

@override
onClose() {
  titleC.dispose();
}

```



```
contentC.dispose();  
}
```

Buatkan fungsi addPost yang digunakan untuk menambahkan data ke dalam sistem melalui service post_service

```
Future addPost() async {  
  update();  
  postModel = await PostApi().addPostAPI(titleC.text, contentC.text);  
  if (postModel!.status == 200) {  
    update();  
    Get.offAndToNamed(Routes.HOME);  
  } else if (postModel!.status == 404) {  
    update();  
  }  
}
```

Berikut kode lengkap untuk add_post_controller.dart

```
class AddPostController extends GetxController {  
  PostModel? postModel;  
  
  RxBool isLoading = false.obs;  
  RxBool isLoadingCreateTodo = false.obs;  
  
  TextEditingController titleC = TextEditingController();  
  TextEditingController contentC = TextEditingController();  
  
  @override  
  onClose() {  
    titleC.dispose();  
    contentC.dispose();  
  }  
  
  Future addPost() async {  
    update();  
    postModel = await PostApi().addPostAPI(titleC.text, contentC.text);  
    if (postModel!.status == 200) {  
      update();  
      Get.offAndToNamed(Routes.HOME);  
    } else if (postModel!.status == 404) {  
      update();  
    }  
  }  
}
```

- Atur file add_post_view.dart menjadi seperti berikut ini :

```
class AddPostView extends GetView<AddPostController> {
  const AddPostView({Key? key}) : super(key: key);
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(
          'Tambah Post',
          style: TextStyle(
            color: AppColor.secondary,
            fontSize: 14,
          ),
        ),
        leading: IconButton(
          onPressed: () => Get.back(),
          icon: SvgPicture.asset('assets/icons/arrow-left.svg'),
        ),
        backgroundColor: Colors.white,
        elevation: 0,
        centerTitle: true,
        bottom: PreferredSize(
          preferredSize: Size.fromHeight(1),
          child: Container(
            width: MediaQuery.of(context).size.width,
            height: 1,
            color: AppColor.secondaryExtraSoft,
          ),
        ),
      ),
      body: ListView(
        shrinkWrap: true,
        physics: BouncingScrollPhysics(),
        padding: EdgeInsets.all(20),
        children: [
          CustomInput(
            controller: controller.titleC,
            label: 'Nama Post',
            hint: 'Berita Terkini',
          ),
          CustomInput(
            controller: controller.contentC,
            label: 'Kontent Post',
            hint: 'Ini contoh content post',
          ),
          SizedBox(height: 32),
          Container(
            width: MediaQuery.of(context).size.width,
            child: Obx(
              () => ElevatedButton(
                onPressed: () {
```



```

RxBool isLoading = false.obs;
RxBool isLoadingCreateTodo = false.obs;

TextEditingController titleC = TextEditingController();
TextEditingController contentC = TextEditingController();

```

Tambahkan fungsi onClose, yang digunakan untuk menghilangkan controller yang sudah di definisikan.

```

@override
onClose() {
  titleC.dispose();
  contentC.dispose();
}

```

Tambahkan fungsi onInit untuk mengisi nilai pada masing-masing widget text sesuai dengan data dari halaman sebelumnya / dari variabel argument

```

@override
void onInit() {
  super.onInit();
  titleC.text = argsData["title"];
  contentC.text = argsData["content"];
}

```

Berikut kode lengkap untuk detail_post_controller.dart

```

class DetailPostController extends GetxController {
  //TODO: Implement DetailPostController

  final Map<String, dynamic> argsData = Get.arguments;
  PostModel? postModel;

  RxBool isLoading = false.obs;
  RxBool isLoadingCreateTodo = false.obs;

  TextEditingController titleC = TextEditingController();
  TextEditingController contentC = TextEditingController();

  @override
  onClose() {
    titleC.dispose();
    contentC.dispose();
  }
}

```

```

@override
void onInit() {
  super.onInit();
  titleC.text = argsData["title"];
  contentC.text = argsData["content"];
}
}

```

- Atur file `detail_post_view.dart` menjadi seperti berikut ini :

```

class DetailPostView extends GetView<DetailPostController> {
  const DetailPostView({Key? key}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(
          'Detail Todo',
          style: TextStyle(
            color: AppColor.secondary,
            fontSize: 14,
          ),
        ),
        leading: IconButton(
          onPressed: () => Get.back(),
          icon: SvgPicture.asset('assets/icons/arrow-left.svg'),
        ),
        actions: [
          TextButton(
            onPressed: () {
              Get.toNamed(Routes.EDIT_POST, arguments: controller.argsData);
            },
            child: Text('Edit'),
            style: TextButton.styleFrom(
              primary: AppColor.primary,
            ),
          ),
        ],
      backgroundColor: Colors.white,
      elevation: 0,
      centerTitle: true,
      bottom: PreferredSize(
        preferredSize: Size.fromHeight(1),
        child: Container(
          width: MediaQuery.of(context).size.width,
          height: 1,
          color: AppColor.secondaryExtraSoft,
        ),
      ),
    ),
  )
}

```

```

),
body: ListView(
  shrinkWrap: true,
  physics: BouncingScrollPhysics(),
  padding: EdgeInsets.all(20),
  children: [
    CustomInput(
      controller: controller.titleC,
      label: 'Nama Post',
      hint: 'Berita Terkini',
      disabled: true,
    ),
    CustomInput(
      controller: controller.contentC,
      label: 'Kontent Post',
      hint: 'Ini contoh content post',
      disabled: true,
    ),
  ],
),
);
}
}

```

- Selesai, silahkan di jalankan dengan menggunakan perintah `flutter run`.

19. Membuat halaman Edit Post

- Ketikkan perintah berikut pada terminal untuk membuat halaman edit post pada Get X:

```
get create page:edit_post
```

- Atur file edit_post_controller.dart menjadi seperti berikut ini :
Definisikan variabel yang akan digunakan untuk mendapatkan data dari halaman sebelumnya dengan menggunakan argument, pemanggilan model post serta definisikan controller untuk widget text

```

PostModel? postModel;
final Map<String, dynamic> argsData = Get.arguments;

RxBool isLoading = false.obs;
RxBool isLoadingCreatePost = false.obs;

TextEditingController titleC = TextEditingController();

```

```
TextEditingController contentC = TextEditingController();
```

Tambahkan fungsi onClose, yang digunakan untuk menghilangkan controller yang sudah di definisikan.

```
@override  
onClose() {  
  titleC.dispose();  
  contentC.dispose();  
}
```

Tambahkan fungsi onInit untuk mengisi nilai pada masing-masing widget text sesuai dengan data dari halaman sebelumnya / dari variabel argument

```
@override  
void onInit() {  
  super.onInit();  
  titleC.text = argsData["title"];  
  contentC.text = argsData["content"];  
}
```

Buatkan fungsi editPost untuk melakukan edit post dengan menggunakan service dari post_service

```
Future<void> editPost() async {  
  update();  
  postModel =  
    await PostApi().editPostAPI(titleC.text, contentC.text, argsData["id"]);  
  if (postModel!.status == 200) {  
    update();  
    Get.offAndToNamed(Routes.HOME);  
  } else if (postModel!.status == 404) {  
    update();  
  }  
}
```

Berikut kode lengkap untuk edit_post_controller.dart

```
class EditPostController extends GetxController {  
  //TODO: Implement EditPostController  
  PostModel? postModel;  
  final Map<String, dynamic> argsData = Get.arguments;  
  
  RxBool isLoading = false.obs;
```

```

RxBool isLoadingCreatePost = false.obs;

TextEditingController titleC = TextEditingController();
TextEditingController contentC = TextEditingController();

@override
onClose() {
  titleC.dispose();
  contentC.dispose();
}

@override
void onInit() {
  super.onInit();
  titleC.text = argsData["title"];
  contentC.text = argsData["content"];
}

Future<void> editPost() async {
  update();
  postModel =
    await PostApi().editPostAPI(titleC.text, contentC.text, argsData["id"]);
  if (postModel!.status == 200) {
    update();
    Get.offAndToNamed(Routes.HOME);
  } else if (postModel!.status == 404) {
    update();
  }
}
}

```

- Atur file `edit_post_view.dart` menjadi seperti berikut ini :

```

class EditPostView extends GetView<EditPostController> {
  const EditPostView({Key? key}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(
          'Edit Post',
          style: TextStyle(
            color: AppColor.secondary,
            fontSize: 14,
          ),
        ),
      ),
      leading: IconButton(
        onPressed: () => Get.back(),

```



```

        icon: SvgPicture.asset('assets/icons/arrow-left.svg'),
      ),
      backgroundColor: Colors.white,
      elevation: 0,
      centerTitle: true,
      bottom: PreferredSize(
        preferredSize: Size.fromHeight(1),
        child: Container(
          width: MediaQuery.of(context).size.width,
          height: 1,
          color: AppColor.secondaryExtraSoft,
        ),
      ),
    ),
  ),
  body: ListView(
    shrinkWrap: true,
    physics: BouncingScrollPhysics(),
    padding: EdgeInsets.all(20),
    children: [
      CustomInput(
        controller: controller.titleC,
        label: 'Nama Post',
        hint: 'Berita Terkini',
      ),
      CustomInput(
        controller: controller.contentC,
        label: 'Kontent Post',
        hint: 'Ini contoh content post',
      ),
      SizedBox(height: 32),
      Container(
        width: MediaQuery.of(context).size.width,
        child: Obx(
          () => ElevatedButton(
            onPressed: () {
              if (controller.isLoading.isFalse) {
                controller.editPost();
              }
            },
            child: Text(
              (controller.isLoading.isFalse) ? 'Edit Post' : 'Loading...',
              style: TextStyle(
                fontSize: 16,
                fontFamily: 'poppins',
              ),
            ),
          ),
        ),
        style: ElevatedButton.styleFrom(
          primary: AppColor.primary,
          padding: EdgeInsets.symmetric(vertical: 18),
          elevation: 0,
        ),
      ),
    ],
  ),
),

```

```

    shape: RoundedRectangleBorder(
      borderRadius: BorderRadius.circular(8),
    ),
  ),
),
),
),
),
],
),
);
}
}

```

- Selesai, silahkan di jalankan dengan menggunakan perintah ``flutter run``.

20. Membuat Delete Post

- Atur file detail_post_controller.dart menjadi seperti berikut ini : (menambahkan fungsi deletePost)

```
Future<void> deletePost() async {
  CustomAlertDialog.showPresenceAlert(
    title: "Hapus data todo",
    message: "Apakah anda ingin menghapus data todo ini ?",
    onCancel: () => Get.back(),
    onConfirm: () async {
      Get.back(); // close modal

      update();
      postModel = await PostApi().deletePostAPI(argsData["id"]);
      if (postModel!.status == 200) {
        update();
        Get.offAndToNamed(Routes.HOME);
      }
    },
  );
}
```

- Tambahkan kode berikut ini pada file detail_post_view.dart dibawah kode CustomInput sehingga akan menambahkan tombol baru yang digunakan untuk mentrigger hapus post :

```
ElevatedButton(
  onPressed: () {
    controller.deletePost();
  },
```

```
child: Text(  
  'Delete post',  
  style: TextStyle(  
    fontSize: 16,  
    fontFamily: 'poppins',  
  ),  
,  
style: ElevatedButton.styleFrom(  
  primary: AppColor.warning,  
  padding: EdgeInsets.symmetric(vertical: 18),  
  elevation: 0,  
  shape: RoundedRectangleBorder(  
    borderRadius: BorderRadius.circular(8),  
  ),  
,  
,  
,  
)
```

- Selesai, silahkan di jalankan dengan menggunakan perintah `flutter run`.